Kernel Affine Hull Features for Time-Series Forecasting: Feature Spans, Bounded Corrections, and Matched Evaluation*

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Abstract

Kernel Affine Hull Machines (KAHMs) reconstruct observations from local reference sets. We study whether these reconstructions provide useful features for a finite-dimensional forecasting decoder. We interpret normalized target transfer as a ratio of kernel-ridge predictions, connecting the established KAHM construction to rescaled kernel interpolation. We give sufficient conditions under which KAHM and matched radial-basis-function (RBF) reconstructions generate the same augmented feature span, and construct an irregular reference set for which the spans differ. Separately, bounded features and a ridge residual decoder yield a uniform bound on the forecast correction, including its free intercept. A prospectively specified experiment compares ten procedures on 192 task suites: three nonlinear simulation laws and a linear control, two observation-noise levels, and 24 independent repetitions. Local dimension is selected before correction calibration, independently for each geometric procedure. Bounded KAHM departure reduces aggregate nonlinear normalized mean squared error by 9.01% relative to common local-coordinate features and 8.01% relative to raw-input RBF forecasting; both establish a prespecified 3% margin under simultaneous inference. Its gains over independently selected RBF departures and boosted trees are 0.72% and 2.39%, below that margin. A selected non-KAHM procedure attains lower error. Dimension selection improves KAHM by 3.26% relative to a fixed cap of two, while a default cap of twenty increases error in these twelve-dimensional histories. Matched-bank controls attribute the dimension improvement to stronger common features despite a smaller incremental departure benefit. Correction bounds and linear-control noninferiority hold, but the joint performance criterion is not established. Supporting studies delimit transfer to natural series and adaptive procedures. The evidence supports conditional forecasting value without establishing a general advantage specific to KAHM.

1 Introduction

A reconstruction of an observed history and a prediction of its future solve different problems. Kernel Affine Hull Machines (KAHMs) construct nonlinear reconstruction maps from local reference observations (Kumar et al., 2024). A small reconstruction error does not establish that the resulting representation is useful for forecasting. For example, a reconstruction residual that vanishes along an affine line loses position along that line, although the future target may vary with position. Conversely, a deterministic feature cannot add information to the history from which it is computed, but it can make useful functions available to a restricted decoder.

This distinction raises three questions. Does the reconstruction supply functions that a simpler matched representation cannot express? Can the magnitude of a fitted forecast correction be controlled? Does the

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resulting procedure improve prediction at the available sample size and computational budget? Each question requires its own evidence. A comparison of function spans concerns algebraic expressiveness; a correction bound concerns the size of one prediction component; a forecasting comparison concerns the complete fitted and selected procedure.

We investigate these questions through *reconstruction departures*. For each local model, a departure is the difference between its nonlinear reconstruction and its affine projection, multiplied by a locality weight. The decoder also receives the locality weights and weighted principal coordinates. These common features retain principal-coordinate position within each local model and are shared by reconstructions fitted to the same bank. Replacing the nonlinear map with its affine projection makes the raw departure zero. A positive-kernel RBF reconstruction, using the same references, kernel, and locality weights, provides a closely matched nonlinear alternative. A residual decoder then adds a calibrated correction to a fixed raw-history ridge forecast.

The algebraic analysis identifies both redundancy and distinctness. Normalized target transfer interprets the established KAHM construction (Kumar et al., 2024) as a ratio of two kernel-ridge predictions, with an antecedent in rescaled kernel interpolation (Deparis et al., 2014; De Marchi et al., 2017). The feature-span comparison asks a separate question. Under a constant-row-sum and invariant-subspace condition, KAHM and RBF departures have the same augmented raw feature span. An irregular three-reference construction shows that the spans can also differ. These results concern untransformed functions. They do not equate fitted ridge predictions under different parameterizations, nor do they establish an advantage for every reference bank or target.

We also analyze the magnitude of the fitted residual correction using the spectral form of centered ridge regression. Each feature block is standardized on the fitting set and mapped into a bounded ball before ridge regression is fitted with an unpenalized intercept. The coefficient and correction bounds account for the empirical residual mean and the scaling of the penalty. They apply to bounded features generally and therefore support the specified KAHM construction without implying a property unique to KAHM. In particular, they do not bound the raw baseline or the total forecast error.

The principal experiment compares ten procedures over four simulation laws, two added-noise levels, and 24 independent repetitions. Fitting, hyperparameter selection, correction calibration, and final evaluation use separate episode sets. Each geometric procedure selects its local dimension cap using B before C calibrates the correction. Fixed-cap and shared-bank comparisons distinguish the complete selection procedure from the contribution of its reconstruction map. The source and analysis were frozen locally before generating the experimental data, all fitted states were fixed before generating final-evaluation episodes, and forecasts were sealed before their targets were released. This was not a public preregistration. Five simultaneous comparisons test a 3% nonlinear-loss improvement margin, and a sixth tests noninferiority to ridge on a linear control with a 5% tolerance.

The bounded KAHM departure establishes the specified improvement over common local-coordinate features and raw RBF forecasting, together with linear-control noninferiority. Its gains over independently selected RBF departures and boosted trees are below the 3% margin, and the selected non-KAHM procedure has lower aggregate loss. Thus three of the six criteria are established, rather than the complete comparison family. Dimension selection improves the complete KAHM procedure relative to a fixed cap of two, but matched-bank decomposition shows a stronger common-feature predictor and a smaller incremental contribution from departures. Increasing the cap to twenty does not improve the aggregate in these twelve-dimensional histories. Supporting controlled, natural-data, and adaptive studies show why this distinction matters beyond the principal experiment. Their findings are summarized in Section 7; their complete methods and results appear in the companion *Supplementary Studies*.

The paper thus contributes sufficient conditions for equality of matched reconstruction feature spans, an explicit example of distinct spans, an intercept-inclusive correction bound for the specified residual decoder, and a dimension-aware evaluation of their forecasting value. These findings separate a useful reconstruction feature from an established advantage specific to KAHM. Full proofs of the central results and the complete principal experimental protocol are included in Appendices A and B.

2 Related work

Local geometry and reconstruction. Prediction from delay coordinates using local approximation predates the present representations (Farmer & Sidorowich, 1987). KAHMs were introduced for geometrically informed classification, including a differential-privacy application (Kumar et al., 2024), and developed for collaborative learning (Kumar et al., 2025). These works establish the reconstruction construction and its geometric properties, including boundedness under their assumptions. The bound here concerns a separately fitted residual forecast. Those classification results do not establish forecasting utility. Local coordinate coding connects localized representations with approximation of smooth functions (Yu et al., 2009). Local tangent coding adds weighted principal-direction displacements (Yu & Zhang, 2010). Our common block has this gate-and-local-coordinate form, with the locality weights and reference banks specified below. With linear readouts these features are also related to local affine experts (Jacobs et al., 1991). Our question is whether the additional KAHM reconstruction departures change the available functions and improve a specified forecasting procedure beyond these common features.

Kernel regression and predictive representations. The matched positive-kernel reconstruction is a vector-valued kernel smoother that averages reference coordinates with normalized Gaussian weights (Nadaraya, 1964). Rescaled kernel interpolation already divides the interpolant of supplied values by the interpolant of the constant function (Deparis et al., 2014; De Marchi et al., 2017). Proposition 1 records the corresponding identity for regularized KAHM weights, while Propositions 2 and 3 characterize equality and inequality of the corresponding augmented raw feature spans. Reconstruction and forecasting objectives also occur in learned dynamical representations: Lusch et al. (2018) combine reconstruction, latent linearity, and multistep prediction, whereas Liu et al. (2023) optimize a residual Koopman forecaster without the canonical reconstruction loss. Here the principal experiment compares complete dimension-selected procedures under common feature-transformation and decoder rules. Secondary comparisons fix the selected geometric bank to compare KAHM and matched RBF departures with the same restricted residual decoder.

Information, regularization, and residual fitting. Supervised dimension reduction studies preservation of the input–target relationship (Fukumizu et al., 2004), and predictive $\mathcal{V}$-information distinguishes information accessible to a restricted observer (Xu et al., 2020). Our feature-space analysis addresses a finite decoder and does not estimate unrestricted Bayes risk on natural data. Residual fitting under squared loss is an established construction (Bühlmann & Yu, 2003). The correction bound in Proposition 4 follows from the singular values of centered ridge regression and is not specific to KAHM. Nyström approximation changes the available coordinates and computational cost (Williams & Seeger, 2001; Rudi et al., 2015); the raw RBF comparator is therefore evaluated as a complete fitted procedure, not as a presumed equivalent function class.

Evaluation and relation to prediction-limit studies. Model-selection bias (Cawley & Talbot, 2010) and forecast-evaluation pitfalls (Hewamalage et al., 2023) motivate distinct fitting, selection, calibration, and evaluation roles. The supporting natural-data studies use the Monash archive (Godahewa et al., 2021) and are reported separately from the principal simulated experiment. Related work by Kumar & Kargaran (2026a) studies successor soft coordinates, norm-bounded minimum-risk certificates, fitted-risk upper bounds, and Koopman closure, and also reports common-target forecasting comparisons. The present study shares KAHM source components and some generator definitions with that work, but forecasts observed future outputs using reconstruction departures and a specified residual decoder. The risk-certificate framework of Kumar & Kargaran (2026b) develops a block hierarchy of lower risk witnesses and certificates of minimum feasible cost among declared representation classes. Here the algebra compares reconstruction feature spans, the deterministic bound controls one fitted correction, and the paired intervals compare complete fitted procedures. They do not certify class-wide impossibility or minimum feasible representation cost.

3 Forecasting features from local reconstruction

Let (y_t) be a scalar observed series. At forecast origin t, the available history is $x_t = (y_{t-d+1}, \ldots, y_t)^\top \in \mathbb{R}^d$, and the target at horizon $h \geq 1$ is y_{t+h}. A forecasting procedure maps the history to a separate prediction for each h in a specified finite horizon set $\mathcal{H}$. The representation and decoder are fitted using a set A of

history–target pairs. Distinct episode sets B, C, and D are used for model selection, correction calibration, and final evaluation, respectively; Section 6 specifies their construction.

Vectors are columns. The notation $v^\top$ denotes transpose, $\|v\|_2$ the Euclidean norm, and $\|v\|_1$ the sum of absolute coordinates. The matrix I_p is the $p \times p$ identity, and $\mathbf{1}_n$ is the length-n vector of ones; omitted subscripts are determined by the displayed products. Expectations and variances refer to the probability law specified for each statement or experiment. All feature-space statements condition on a fixed fitted reference bank.

3.1 Reference banks and reconstruction maps

Let $z \in \mathbb{R}^d$ be the coordinatewise standardized version of a history x, using means and scales estimated on A. A local reference bank is a collection of fitted models indexed by j. Model j has n_j reference histories stored as the rows of $Q_j \in \mathbb{R}^{n_j \times d}$, center $\mu_j \in \mathbb{R}^d$, and orthonormal principal directions $P_j \in \mathbb{R}^{d \times r_j}$. Here r_j is its local rank and $P_j^\top P_j = I_{r_j}$. Write $q_{ji} \in \mathbb{R}^d$ for reference i, so row i of Q_j is $q_{ji}^\top$. Let S_j be the sample covariance matrix of the projected references $P_j^\top q_{ji}$. For positive rank and positive-definite S_j, the shared Gaussian kernel is

$$\kappa_j(z, z') = \exp\left\{ -\frac{[P_j^\top (z - z')]^\top S_j^{-1} [P_j^\top (z - z')]}{2r_j} \right\}.$$

Define the kernel vector $k_j(z) \in \mathbb{R}^{n_j}$ by $[k_j(z)]_i = \kappa_j(z, q_{ji})$, and the Gram matrix $K_j \in \mathbb{R}^{n_j \times n_j}$ by $[K_j]_{i\ell} = \kappa_j(q_{ji}, q_{j\ell})$. With a positive reconstruction regularizer λ_j, set $C_j = (K_j + \lambda_j I_{n_j})^{-1}$. The Gaussian Gram matrix and its regularized inverse C_j are symmetric. The regularizer is fitted from A by the reference-reconstruction rule specified in Appendix B.

The KAHM and matched positive-kernel reconstructions are

$$A_{K,j}(z) = \frac{Q_j^\top C_j k_j(z)}{\mathbf{1}^\top C_j k_j(z)}, \qquad A_{R,j}(z) = \frac{Q_j^\top k_j(z)}{\mathbf{1}^\top k_j(z)}. \tag{1}$$

Both take values in $\mathbb{R}^d$. The first is defined when its signed denominator is nonzero; the second has a positive denominator for finite inputs. The matched affine reconstruction is $A_{A,j}(z) = \mu_j + P_j P_j^\top (z - \mu_j)$.

Local dimension. For a supplied cap $c \geq 1$, the native construction initializes the local dimension at

$$r_j^{\text{init}} = \min\{c, d, n_j - 1\}.$$

The default cap is $c = 20$ (Kumar et al., 2025, Appendix B). The stored rank r_j can be smaller because the native PCA retains only positive-variance directions and reduces the requested dimension when a projected reference coordinate has insufficient range. The range rule stops at requested rank one; it does not impose the range threshold on every possible rank-one model. Appendix B gives the exact numerical rules and the explicit rank-zero wrapper. There is no additional $d - 1$ cap in this native policy. In particular, full ambient rank is permitted, and a cap of twenty does not imply twenty stored directions when $d < 20$ or $n_j \leq 20$.

3.2 Common features and reconstruction departures

Models are grouped into partitions. If $\mathcal{J}_b$ indexes the models in partition b, their shared locality weights are

$$g_j(z) = \frac{\exp\{-\|z - \mu_j\|_2^2/(2s_j)\}}{\sum_{\ell \in \mathcal{J}_b} \exp\{-\|z - \mu_\ell\|_2^2/(2s_\ell)\}}, \qquad j \in \mathcal{J}_b,$$

where $s_j > 0$ is the fitted local squared-radius scale. These weights, called gates, sum to one within each partition and do not depend on the reconstruction being compared.

The common feature vector $\psi_0(z)$ contains all gates $g_j(z)$ and all weighted principal-coordinate vectors $g_j(z) P_j^\top (z - \mu_j)$. Define the reconstruction departures by

$$\begin{aligned} d_K(z) &= \text{concat}_j \big[g_j(z) \{ A_{K,j}(z) - A_{A,j}(z) \} \big], \\ d_R(z) &= \text{concat}_j \big[g_j(z) \{ A_{R,j}(z) - A_{A,j}(z) \} \big]. \end{aligned} \tag{2}$$

The operator concat stacks its column-vector arguments in a fixed order. Each departure retains all history coordinates, including components outside the local principal subspace. Substituting the affine map for the nonlinear reconstruction makes the departure identically zero. The common-feature procedure is therefore the nested control without reconstruction departures. Within a fixed bank, the RBF departure is matched to the KAHM departure in references, kernels, gates, and feature dimension.

The affine control is nested at the level of the raw feature construction. The comparison with positive-kernel RBF reconstruction additionally holds fixed the number and locations of references, the projected kernel, the locality weights, and the decoder architecture. The difference between the two raw departure blocks is therefore the use of C_j in the normalized reconstruction. This matching does not assert equality of fitted predictions, and it does not compare KAHM with every possible kernel or nonlinear decoder. When complete procedures select their caps independently, their selected banks and common-feature dimensions can differ. We therefore distinguish those primary forecasting comparisons from secondary comparisons that hold the KAHM-selected bank fixed.

3.3 Defined evaluation of signed normalizations

To guard against cancellation in the signed KAHM normalization, define

$$a_j(z) = C_j k_j(z), \qquad \rho_j(z) = \frac{|\mathbf{1}^\top a_j(z)|}{\|a_j(z)\|_1} \quad \text{when } \|a_j(z)\|_1 > 0. \tag{3}$$

The ratio is invariant to a common positive rescaling of the query kernel. When $\rho_j(z) \le 10^{-10}$, or a required quantity is undefined or nonfinite, the local reconstruction is replaced by its affine map, making that raw departure zero. The same finite-value checks apply to the positive RBF map. All affected rows are recorded. This fallback gives a defined feature vector where the original ratio cannot be evaluated reliably; it does not establish continuity across the fallback boundary.

A zero raw departure under fallback need not remain zero after A-fitted feature centering. The fallback defines the input to the subsequent transformation; the norm bound in Section 5 applies to that transformed input. The algebraic propositions in Section 4 concern the exact ratios on their stated domains, rather than the piecewise numerical fallback procedure.

4 Normalized reconstruction and feature spans

The reconstruction formulas permit two different questions: whether a particular map is an established kernel predictor, and whether its coordinates give a linear decoder functions unavailable from the matched reconstruction. We answer these questions for fixed reference geometry. The results concern exact functions before feature standardization, radial transformation, or penalized fitting.

Consider one local model from Section 3 and suppress its index. Write M for its number of references, $Q \in \mathbb{R}^{M\times d}$ for their row-stored coordinates, and $C = (K + \lambda I_M)^{-1}$ for its regularized inverse, with $\lambda > 0$. The matrix K is the symmetric positive-semidefinite Gaussian Gram matrix. The vector $\mathbf{1}_M$ contains M ones, and $k(z) \in \mathbb{R}^M$ contains the kernel similarities of a query z to the references. Let A_K, A_R, and A_A denote the KAHM, positive-kernel RBF, and affine reconstructions in Equation (1).

4.1 The meaning of normalized target transfer

Proposition 1 (Normalization and target transfer)**.** *For a query with $s(z) = \mathbf{1}_M^\top C k(z) \neq 0$, define $\omega(z) = Ck(z)/s(z)$. Then $\mathbf{1}_M^\top \omega(z) = 1$, and for every vector of reference targets $y \in \mathbb{R}^M$,*

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1} \mathbf{1}_M}. \tag{4}$$

This is the ratio of two kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept. For an affine scalar function $f(q) = q^\top b + c$, where $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, its reference values $f(Q) = Qb + c\mathbf{1}_M$ satisfy

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

The proof uses symmetry of C and the fact that the normalized weights sum to one. The numerator in Equation (4) fits the supplied targets, while the denominator fits the constant target. Because the denominator varies with the query, the ratio generally differs from the ordinary unnormalized kernel-ridge predictor. The affine identity also specifies the effect of reconstruction error on an affine target: only its component in the target direction b enters. The complete proof is in Appendix A.

The quotient normalization has an established interpolation antecedent. When K is invertible, setting $\lambda = 0$ in the quotient gives the rescaled kernel interpolant of De Marchi et al. (2017), wherever its denominator is nonzero. Proposition 1 interprets the existing KAHM weights with positive regularization; it does not introduce a new normalization principle.

4.2 Equal spans under reference symmetry

For a fixed common gate $g(z)$, define $d_K(z) = g(z)\{A_K(z) - A_A(z)\}$ and $d_R(z) = g(z)\{A_R(z) - A_A(z)\}$. Let $\mathcal{S}_0$ be the linear span of the scalar gate and the coordinates of $g(z)P^\top(z-\mu)$, where μ and P are the reference center and principal directions. The coordinates of $g(z)A_A(z) = g(z)\mu + P\{g(z)P^\top(z-\mu)\}$ belong to $\mathcal{S}_0$. Adjoining a vector-valued feature to a span means adjoining each of its scalar coordinate functions. A common intercept may be adjoined to both spans without changing the equality below.

Because gA_A already belongs coordinatewise to $\mathcal{S}_0$, adjoining a departure gives the same raw span as adjoining the corresponding gated reconstruction. Subtracting A_A specifies a parameterization; the span comparison concerns the additional functions supplied by the nonlinear map.

Proposition 2 (A sufficient condition for equal feature spans)**.** *Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ, and C leaves $\mathcal{V} = \operatorname{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\}$ invariant. Here $Q_{:i}$ is column i of Q, and invariance means $C\mathcal{V} \subseteq \mathcal{V}$. Then*

$$\operatorname{span}(\mathcal{S}_0, d_K) = \operatorname{span}(\mathcal{S}_0, d_R).$$

More explicitly, there are $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that, with $a = (\kappa+\lambda)^{-1} > 0$,

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad A_K(z) = \{B^\top A_R(z) + c\}/a.$$

Invariance supplies the displayed representation of CQ. Substitution in the normalized reconstruction expresses each KAHM departure using RBF departures and the common affine features. The restriction of C to the finite-dimensional space $\mathcal{V}$ is invertible onto that space, which gives the reverse inclusion. This argument also covers redundant reference-coordinate columns; it does not require B to be invertible. Appendix A gives the full proof.

A two-reference example makes the role of the common coordinates explicit. Take scalar references $(-1, 1)$, $g = 1$, $P = 1$, and $A_A(z) = z$. If ρ is the off-diagonal Gram entry, then

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1+\rho+\lambda}{1-\rho+\lambda} > 0.$$

The departures differ, but adjoining the coordinate z makes their augmented spans equal. Thus a nonlinear reconstruction formula alone does not establish an additional raw basis.

4.3 An irregular reference bank gives a distinct basis

Proposition 3 (Irregular scalar references)**.** *Let $q_1 < q_2 < q_3$ satisfy $q_1 + q_3 \neq 2q_2$, and let $k_i(z) = \exp\{-t(z-q_i)^2/2\}$ for a precision $t > 0$. Set $K_{ij} = \exp\{-t(q_i-q_j)^2/2\}$ and $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$. Suppose every entry of $C\mathbf{1}_3$ is positive. With $Q = (q_1, q_2, q_3)^\top$, $g = 1$, and $A_A(z) = z$,*

$$A_K \notin \operatorname{span}(1, z, A_R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

These spans consist of scalar functions on the whole real line.

The six products of kernel functions have distinct exponential factors because their exponents contain the six distinct sums $q_i + q_j$ for $i \leq j$. Their linear independence rules out an affine relation between A_K and

A_R: such a relation would force K to have constant row sums, whereas its middle row has a strictly larger sum than either outer row. Positivity of $C\mathbf{1}_3$ makes both reconstructions bounded, excluding an additional nonzero multiple of z. Appendix A provides every step of this argument.

For the explicit references $(-1, 0, 2)$, the sample-variance precision is $t = 3/7$. The reference-based regularization rule gives $\lambda = 4.498274743339644$, for which positivity of $C\mathbf{1}_3$ follows from the matrix bound in the appendix. This supplies a concrete example satisfying the proposition. It does not assert distinctness for every higher-dimensional bank fitted in the forecasting experiment.

Together, the propositions delimit the algebraic contribution. Some reference geometries yield equal augmented spans, while an irregular geometry can supply a distinct function. Neither conclusion determines forecasting accuracy. A general change of feature coordinates changes an isotropic ridge penalty, and separate nonlinear radial transformations can remove a raw linear span relation. The empirical comparison therefore evaluates the complete feature transformation and fitted decoder as well as the reconstruction map.

5 Bounded residual forecasting

5.1 Separate bounded feature blocks

Each feature block is centered and scaled coordinatewise using A alone; feature standard deviations are floored at 10^{-6}. A tilde denotes this coordinatewise standardization. For a vector $v \in \mathbb{R}^p$, define

$$T_R(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R^2}}, \qquad R = \sqrt{p}. \tag{5}$$

This radial map has norm at most R and is nonexpansive with respect to its argument. If the common and departure block dimensions are p_0 and p_d, respectively, set $R_0 = \sqrt{p_0}$ and $R_d = \sqrt{p_d}$. The bounded feature vectors are

$$\psi_K(z) = \mathrm{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_K(z))\big), \qquad \psi_R(z) = \mathrm{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_R(z))\big).$$

The two blocks are transformed separately and are not standardized again. Consequently, appending a departure does not rescale the common features. For J local models with stored ranks r_j, the dimensions are $p_0 = J + \sum_j r_j$ and $p_d = Jd$. The principal experiment requests partitions with 2, 4, and 8 models and at most 24 references per model. All fitted banks contain $J = 14$ models, so $p_d = 168$ for its 12-dimensional histories, whereas p_0 varies with the local ranks. Each candidate geometry is fitted without targets on A and remains fixed during selection and evaluation. All bounds use its actual dimensions.

5.2 Fitting and calibrating the correction

For horizon h, let $f_{0,h}$ be the raw-history ridge baseline. Given an unstandardized history x, let $\psi(x)$ denote the chosen bounded feature vector after applying its A-fitted input scaler. The forecast is

$$\widehat{y}_{t+h} = f_{0,h}(x_t) + \gamma_h q_h(x_t), \qquad q_h(x) = b_h + \psi(x)^\top w_h. \tag{6}$$

Here $b_h \in \mathbb{R}$ is a free intercept, w_h is the coefficient vector, and γ_h controls the correction strength. For n_A fitting histories, the residual targets are $e_{ih} = y_{i,h} - f_{0,h}(x_i)$, where $y_{i,h}$ is the observed future at horizon h for fitting history i. The correction minimizes

$$\frac{1}{n_A}\sum_{i=1}^{n_A}\{e_{ih} - b_h - \psi(x_i)^\top w_h\}^2 + \alpha_h\|w_h\|_2^2.$$

Set B selects $\alpha_h \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ by forecast error separately for each horizon and candidate cap. For each geometric procedure m, let $\widehat{y}^{(c)}_{m,h}$ be its B prediction after this penalty selection, with $\gamma_h = 1$. One cap is selected across horizons by

$$\widehat{c}_m = \operatorname*{arg\,min}_{c\in\{1,2,4,8,12,20\}} \frac{1}{|\mathcal{H}|}\sum_{h\in\mathcal{H}} \frac{\mathrm{MSE}_B(Y_h, \widehat{y}^{(c)}_{m,h})}{\max\{\mathrm{Var}_A(Y_h), 10^{-12}\}}. \tag{7}$$

Here Var_A is the empirical variance of the retained A targets with divisor n_A; exact ties select the smallest cap. The common, bounded KAHM, bounded RBF, and two unbounded-departure procedures each make their own choice. These choices are fixed before accessing C. Set C then selects $\gamma_h \in \{0, 1/4, 1/2, 1\}$ separately by horizon. Neither selection step refits the geometry or decoder, and set D is reserved for final evaluation. The cap objective uses uncalibrated B forecasts, so C does not enter dimension selection.

5.3 A uniform bound including the intercept

Define the fitting residual mean, centered root mean square, and feature mean as

$$\bar{e}_h = \frac{1}{n_A}\sum_i e_{ih}, \qquad \sigma_{e,h} = \left[\frac{1}{n_A}\sum_i (e_{ih} - \bar{e}_h)^2\right]^{1/2}, \qquad \bar{\psi} = \frac{1}{n_A}\sum_i \psi(x_i).$$

Proposition 4 (Centered ridge correction bound)**.** *For the mean-squared-error objective above with $n_A \geq 1$, $\alpha_h > 0$, and an unpenalized intercept, suppose that the feature map satisfies $\|\psi(x)\|_2 \leq R$ on its input domain. For any scalar γ_h, the fitted coefficients and correction satisfy*

$$\|w_h\|_2 \leq \frac{\sigma_{e,h}}{2\sqrt{\alpha_h}}, \qquad |\gamma_h q_h(x)| \leq |\gamma_h| \left[|\bar{e}_h| + (R + \|\bar{\psi}\|_2)\frac{\sigma_{e,h}}{2\sqrt{\alpha_h}}\right]. \tag{8}$$

For a concatenated bounded common and departure vector, $R = \sqrt{R_0^2 + R_d^2}$; for the common-only vector, $R = R_0$. The free intercept contributes the residual-mean term and cannot be omitted. This bound controls the correction, not the baseline, total forecast error, or sensitivity of the preceding reconstruction to changes in the history.

Proof idea. Eliminating the intercept gives $b_h = \bar{e}_h - \bar{\psi}^\top w_h$. If Ψ_c is the matrix whose rows are $(\psi(x_i) - \bar{\psi})^\top$ and e_c is the centered residual vector, then $w_h = (\Psi_c^\top \Psi_c + n_A\alpha_h I)^{-1}\Psi_c^\top e_c$. The coefficient operator has singular values $s/(s^2 + n_A\alpha_h) \leq 1/(2\sqrt{n_A\alpha_h})$. This yields the coefficient bound. Applying Cauchy–Schwarz to $q_h(x) = \bar{e}_h + (\psi(x) - \bar{\psi})^\top w_h$ yields the correction bound. Appendix A gives the complete argument, including the nonexpansiveness of the radial map.

The proposition applies separately to each selected horizon and is a property of bounded-feature ridge regression. A summed-squared-error objective would need penalty $n_A\alpha_h$ to define the same estimator. The radial transformation is nonexpansive in its own input, but this does not give a Lipschitz bound for the preceding signed reconstruction or for its fallback boundary.

Baseline recovery and predictive gain. The calibration grid includes $\gamma_h = 0$, which recovers the same fitted baseline exactly. This is an available forecast rather than a guarantee that a finite calibration set selects the best future forecast. The coefficient and correction bounds likewise make no claim about prediction accuracy. The experiment therefore evaluates error, numerical coverage, and the correction inequalities as separate outcomes.

6 Dimension selection and forecasting comparisons

6.1 Comparison procedures

The primary candidate is the bounded KAHM departure. Its five primary comparators are the common features, bounded RBF departure, raw-history RBF forecasting, histogram gradient boosting, and a validation-selected non-KAHM procedure. Raw ridge, persistence, and KAHM and RBF variants that omit only the departure bound complete the ten-method comparison. The principal experiment uses the bounded features and candidate geometries defined in Sections 3 and 5. All non-ridge candidates are calibrated on C by blending their forecasts with the same ridge baseline. The selection policy chooses one complete, calibrated non-KAHM procedure using horizon-averaged normalized C error; neither KAHM variant is eligible. Raw

Table 1: Data roles in each principal task suite. Each episode has 72 observations. A history contains 12 observations and predicts horizons 1, 4, and 12. The final evaluation set is not used to choose any model setting or correction weight.

Set	Episodes	Histories	Role
A	64	512	Fit scalers, reference geometry, baseline, and candidate decoders.
B	32	256	Choose penalties per horizon and one dimension cap per geometric procedure.
C	32	256	Choose correction weights and one complete non-KAHM procedure.
D	128	1,664	Evaluate fixed forecasts and compute the reported losses.

RBF and boosting have the same A/B/C roles and horizon-specific hyperparameter selection. Appendix B specifies their grids and costs.

The primary comparison uses each geometric procedure's independently B-selected cap from Equation (7). Three secondary designs isolate the role of that choice. First, a fixed cap of two uses the explicit restriction $\min\{2, d-1, n_j-1\}$. Second, the native default cap twenty uses $\min\{20, d, n_j-1\}$. Third, all reconstruction controls use the bounded KAHM procedure's selected bank, while retaining their own penalty and calibration choices. Only the third design, and comparisons within a fixed-cap bank, match references, projected kernels, gates, feature dimensions, transformation rules, and decoder grids. Primary independently selected procedures need not share a bank.

All six native caps and the fixed-cap-two bank are fitted separately. At history dimension twelve, native caps twelve and twenty produce identical banks and forecasts; ascending tie resolution therefore never selects twenty. The fixed-cap-two and native-cap-two results are also identical in this experiment. These identities are verified before final evaluation and retained as internal controls. The stored common-feature width ranges from 28 to 182 across candidates; either departure adds 168 coordinates. Thus dimension selection changes the common feature block as well as the projected kernel and reconstruction departure. Raw RBF forecasting instead uses 12 history coordinates and at most 64 Nyström coordinates. Its comparison, and those with boosting and the selection policy, assess complete procedures with different representations and search spaces. They are not feature- or computation-matched ablations.

6.2 Observation laws and independent evaluation roles

The experiment comprises four laws, two additional-noise levels, and 24 independent repetitions, giving 192 task suites. The nonlinear laws are an autoregression, Van der Pol dynamics, and Duffing dynamics. A stable linear Gaussian AR(2) process provides a negative control. Additional Gaussian observation noise has standard deviation ηs_A, where s_A is the standard deviation of all original A episode values and $\eta \in \{0, 0.1\}$. The added noise affects histories and targets. Underlying episodes, standardized noise draws, retained rows, and fitting seeds are paired between the two noise levels. The oscillator observations already contain sensor noise when $\eta = 0$.

Sets A, B, C, and D contain 64, 32, 32, and 128 episodes, respectively, each of length 72. Their base episodes and standardized noise draws use independent random streams; the additional-noise amplitude is estimated on A and shared across sets. Histories have length 12, horizons are $\mathcal{H} = \{1, 4, 12\}$, and consecutive forecast origins are four steps apart. A, B, and C retain at most 512, 256, and 256 histories, respectively; D retains all 1,664 origins. The protocol, executable source, and analysis were frozen before generating these data. B choices are saved before C calibration; all fitted states are sealed before generating D. D inputs and targets are stored separately, and all forecasts are sealed before target release. Fresh random streams are disjoint from the earlier studies' streams. This is a prospective local specification informed by those studies, rather than public preregistration or external replication. Their losses are not pooled with this experiment.

6.3 Loss aggregation and simultaneous inference

For task τ, method m, and horizon h, let $\mathrm{MSE}_{D,\tau,m,h}$ be the mean squared error over D origins. Define $v_{\tau h} = \max\{\mathrm{Var}_{A,\tau}(Y_h), 10^{-12}\}$, where Y_h denotes the horizon-h target and the variance uses the retained A target rows with divisor n_A. The task loss is

$$L_{\tau m} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{\mathrm{MSE}_{D,\tau,m,h}}{v_{\tau h}}.$$

For repetition r, let L^{NL}_{rm} average the six nonlinear law–noise task losses equally, and let L^{lin}_{rm} average the two linear-control task losses. For each of the five nonlinear comparators, define the useful-improvement contrast

$$d_{rm} = 0.97 L^{\mathrm{NL}}_{rm} - L^{\mathrm{NL}}_{rK}.$$

A positive population mean of this contrast corresponds to an improvement in expected normalized loss exceeding the specified 3% margin. The linear-control contrast is

$$d_{r,\mathrm{lin}} = 1.05 L^{\mathrm{lin}}_{r,\mathrm{ridge}} - L^{\mathrm{lin}}_{rK},$$

which tests noninferiority with a 5% relative tolerance. For each contrast, $\bar{d} = 24^{-1} \sum_r d_r$ and $s_d^2 = 23^{-1} \sum_r (d_r - \bar{d})^2$ denote the sample mean and variance. The one-sided lower confidence bound is

$$\bar{d} - t_{23,1-0.05/6} \frac{s_d}{\sqrt{24}},$$

where $t_{\nu,q}$ is quantile q of the Student t distribution with ν degrees of freedom. Bonferroni adjustment covers the six contrasts simultaneously at nominal level 95%, subject to the paired-mean t approximation. The sampling units are independent repetition aggregates; overlapping windows and paired noise levels do not increase the inferential sample size.

The prespecified criterion requires all six lower bounds to exceed zero. It also requires complete finite predictions, satisfaction of the feature, coefficient, and correction bounds, and a taskwise maximum absolute KAHM error no greater than five times the corresponding ridge maximum. The last requirement is an engineering diagnostic rather than a population risk guarantee. A wholly failed method or a final prediction-cell recovery prevents an efficacy conclusion. The 3% improvement margin, 5% tolerance, and fixed 24-repetition budget express the study's research criteria; they do not derive from a deployment loss function or a power calculation. Failure to satisfy the complete criterion concludes the planned performance evaluation without adding repetitions or selecting favorable conditions after inspecting D.

6.4 Predictive results

Table 2 gives the complete ten-procedure comparison for the primary selected context. Bounded KAHM departure has nonlinear mean loss 0.028912, compared with 0.031775 for common features, 0.029122 for bounded RBF departure, 0.031429 for raw RBF, 0.029619 for boosting, and 0.027778 for the selected non-KAHM procedure. The respective point gains are 9.01%, 0.72%, 8.01%, 2.39%, and −4.08%.

The simultaneous lower bounds establish the 3% margins over common features and raw RBF, and the 5% linear-control noninferiority criterion (Table 3). They do not establish the other three margins. In particular, the positive point improvement over boosting does not reach 3%, and the selected non-KAHM procedure has lower mean loss than KAHM. Three of the six criteria are established; the complete joint criterion is not established. Failure to establish an improvement margin does not establish equality of the procedures.

Figure 1 shows every law and noise level for the independently selected RBF reconstruction and the selected non-KAHM procedure. These descriptive comparisons show how the relative error varies across the declared conditions. The appendix gives all ten methods, and the accompanying data include every horizon-specific result.

Table 2: All ten primary procedures after independent B selection and C calibration. Entries average A-variance-normalized D MSE over 24 repetitions and, within each panel, equally over its law–noise conditions and three horizons. The nonlinear and linear panels have different loss scales and are not pooled for inference. "Unbounded" removes only the departure's radial bound. Lower is better.

Procedure	Nonlinear SMSE	Linear SMSE
Raw ridge	0.059223	0.901659
Raw RBF	0.031429	0.904691
Boosted trees	0.029619	0.904476
Persistence	0.059406	0.902609
Common features	0.031775	0.902291
Bounded KAHM departure	0.028912	0.903418
Bounded RBF departure	0.029122	0.902816
Unbounded KAHM departure	0.029414	0.903481
Unbounded RBF departure	0.029436	0.903145
Selected non-KAHM	0.027778	0.906497

Table 3: The six primary comparisons for bounded KAHM departure. L_K and L_m are losses averaged over repetitions. The first five rows use the nonlinear aggregate and $\bar{d} = 0.97L_m - L_K$; the last uses the linear aggregate and $\bar{d} = 1.05L_m - L_K$. Gain is $100(1 - L_K/L_m)$, so negative values favor the comparator. The lower bound concerns the absolute margin contrast, not the percentage gain. All bounds use 24 independent repetitions, 23 degrees of freedom, and a one-sided Bonferroni adjustment for six comparisons. A criterion is satisfied only when its lower bound is positive.

Comparator	L_K	L_m	Gain (%)	$\bar{d}$	Lower bound	Established
Common features	0.02891	0.03178	9.01	0.001910	0.001529	Yes
Bounded RBF departure	0.02891	0.02912	0.72	-0.000663	-0.000800	No
Raw RBF	0.02891	0.03143	8.01	0.001574	0.001156	Yes
Boosted trees	0.02891	0.02962	2.39	-0.000181	-0.000662	No
Selected non-KAHM	0.02891	0.02778	-4.08	-0.001967	-0.002218	No
Raw ridge (linear NI)	0.90342	0.90166	-0.20	0.043324	0.041880	Yes

6.5 Dimension changes the complete feature representation

Table 4 compares the specified cap rules on the same episodes. B-selected KAHM improves on fixed cap two by 3.26%, with lower repetition-average loss in 21 of 24 repetitions. Its mean paired difference is -0.000973 with a secondary, unadjusted 95% paired-t interval $[-0.001229, -0.000717]$. Native cap twenty instead increases loss by 8.39%, with higher loss in all 24 repetition aggregates; its corresponding difference is 0.002507 with interval $[0.002024, 0.002991]$. These intervals are descriptive secondary analyses outside the six-comparison primary family.

The bounded KAHM cap counts across the 144 nonlinear tasks are 15, 22, 68, 31, and 8 for caps 1, 2, 4, 8, and 12, respectively. Cap four is the most frequent selection, but the variation does not support a universal fixed-cap recommendation. No experiment here measures a binding cap of twenty in an ambient dimension greater than twenty.

To interpret the selected procedure, let $L_K^{(c)}$ and $L_0^{(c)}$ denote KAHM and common-feature loss using the same bank rule c, each with its own fitted head and C calibration. Relative to the fixed cap-two bank, the exact loss decomposition is

$$L_K^{(c)} - L_K^{(2)} = \{L_0^{(c)} - L_0^{(2)}\} + \{(L_K^{(c)} - L_0^{(c)}) - (L_K^{(2)} - L_0^{(2)})\}.$$

For the KAHM-selected bank, the complete change -0.000973 comprises a common-feature improvement of -0.002150 and a weakening of the incremental departure benefit by $+0.001177$. The departure still reduces loss relative to this matched common control: its increment changes from -0.004967 to -0.003790. Native

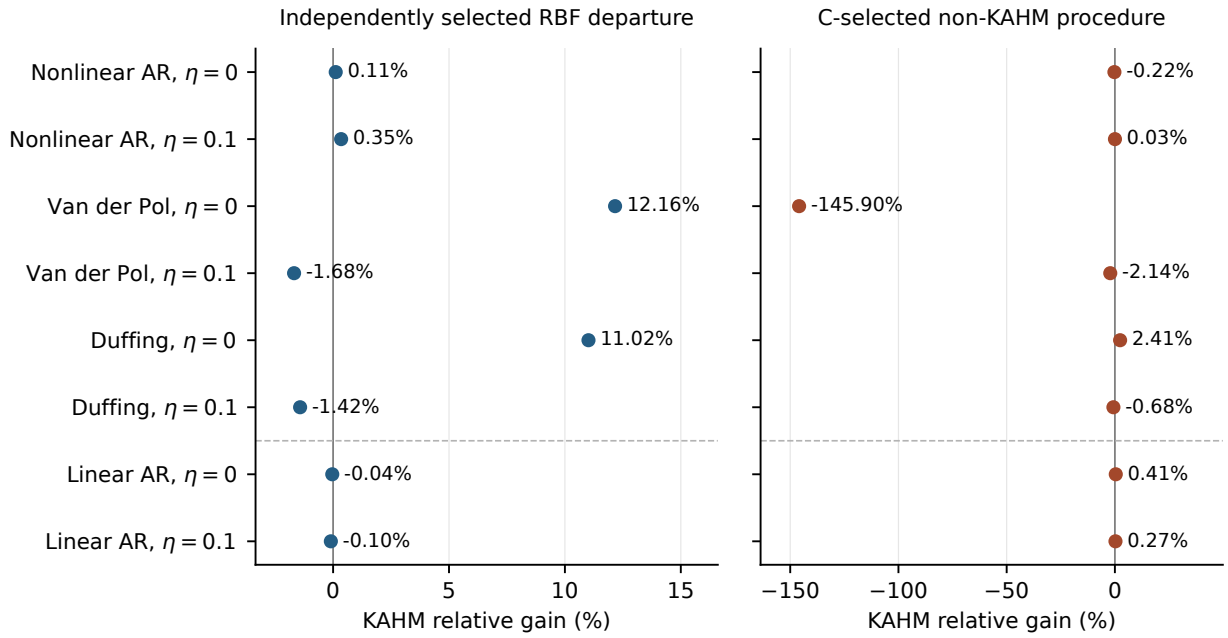


Figure 1: Condition-specific gains of bounded KAHM departure over the independently selected bounded RBF departure and the C-selected non-KAHM procedure. Each point is $100(1 - \overline{L}_K/\overline{L}_m)$, where the bars denote mean loss over 24 repetitions of that condition. Positive values favor KAHM. The panels use different horizontal scales, and labels give the point estimates. These descriptive estimates have no condition-specific confidence claims. The dashed divider separates nonlinear laws from the linear negative control.

Table 4: Nonlinear mean SMSE under fixed and selected dimension rules. Within each fixed row, the three procedures share a bank. In the final row, each selects its own cap on B. The last column compares that row's KAHM loss with fixed cap two. Native caps twelve and twenty are exact aliases because the history dimension is twelve.

Dimension rule	KAHM	Common	RBF	KAHM gain (%)
Fixed cap 2	0.029885	0.034852	0.030557	0.00
Native cap 1	0.030766	0.036284	0.030852	-2.95
Native cap 2	0.029885	0.034852	0.030557	0.00
Native cap 4	0.029780	0.032353	0.029927	0.35
Native cap 8	0.031139	0.032465	0.030477	-4.20
Native cap 12	0.032392	0.033046	0.031417	-8.39
Native cap 20	0.032392	0.033046	0.031417	-8.39
B-selected cap	0.028912	0.031775	0.029122	3.26

cap twenty similarly strengthens the common predictor but loses more of the departure's incremental benefit, producing a net deterioration. This decomposition concerns the complete fitted predictors; it is not an orthogonal variance decomposition or a causal attribution to individual coordinates.

Using the KAHM-selected bank, bounded KAHM improves over matched bounded RBF by 0.68%, compared with 0.72% for independently selected RBF. The matched difference is -0.000198, with secondary unadjusted interval $[-0.000325, -0.000072]$. Its sign does not establish the primary 3% margin. Bounded KAHM also has 2.28% lower loss than its unbounded counterpart on the same bank, versus 1.71% when those procedures select their caps independently. Setting only the calibration weights to one retains a 3.39% selected-versus-fixed-cap-two improvement; the direction of the dimension result therefore does not depend on C shrinkage alone.

The heterogeneity is substantial. Selection improves fixed-cap-two KAHM in five of the six nonlinear law–noise cells, but slightly worsens Duffing with added noise. At Van der Pol with no additional Gaussian noise, KAHM has loss 0.009347 versus 0.003801 for boosting; the non-KAHM policy selects boosting in all 24 repetitions of this condition. Native cap twenty improves the first two horizon averages relative to cap two but worsens horizon twelve. Appendix B reports every method, condition, and horizon aggregate.

6.6 Numerical checks and measured resources

The maximum taskwise KAHM-to-ridge maximum-error ratio is 1.4673, below the limit of five. There are no feature-bound, coefficient-bound, or bounded-correction violations. All 192 tasks complete without a failed method or recovered final forecast. Across 18,816 local models, no model requires an additional positive-eigenvalue or projected-range rank reduction, a rank-zero fallback, or a numerical query guard. Native cap twenty stores rank twelve in 2,685 of its 2,688 local models; three two-reference models have rank one. The stacked local bases have ambient rank twelve for every bank, including the small-cap banks. C selects a nonzero KAHM correction weight in 431 of 432 nonlinear task–horizon combinations and 52 of 144 linear-control combinations. This activation frequency describes calibration choices, not a guarantee of lower D loss. Mean charged fitting and selection time is 0.490 seconds for selected bounded KAHM departure and 0.380 seconds for selected bounded RBF departure. These costs include all six native cap candidates. The fixed-cap-two and native-cap-twenty KAHM costs are 0.085 and 0.101 seconds. Appendix B reports retained state sizes, full-suite timing scope, and the accounting boundaries.

The algebraic example, deterministic bound, and empirical comparisons answer different questions. A distinct feature can be poorly aligned with the target or difficult to estimate from the available sample. A bounded correction can remain inaccurate. An unresolved improvement margin does not establish equality of the representations. The conclusions therefore apply to the specified reconstructions, decoder, selection rules, and observation laws.

7 Supporting evidence and limits of transfer

Complementary studies examine the information retained by geometric features, access by finite decoders, transfer to natural series, and learning or updating the geometry. They use different representations, baselines, splits, loss scales, and sampling units from the bounded reconstruction-departure experiment. Their outcomes therefore describe separate fitted procedures; they are not pooled into its confidence family. Their local-rank caps are the explicitly constrained settings specified in their own protocols, rather than evaluations of the native default-cap or dimension-selected procedures in Section 6. Their evidence cannot be transferred to those procedures by changing the interpretation of the cap. Table 5 summarizes the evidence. The companion *Supplementary Studies* document provides the complete definitions, proofs, protocols, comparison tables, and failure accounting.

Information and finite-decoder access. In a controlled branch construction, the same observations and noise support targets that vary from branch identity to signed position. An affine reconstruction can be nearly exact while its discrepancy-based code loses the position required for prediction. Affine local coordinates recover accurate prediction in that example, so the benefit is not exclusive to nonlinear reconstruction. In a separate finite-decoder comparison, KAHM local features outperform raw and affine local features on Van der Pol at both 64 and 256 Gaussian landmarks. Raw histories perform better on nonlinear autoregression, and both controls perform better on Duffing. The inferential units are 24 independent full repetitions per setting, with separate multiplicity adjustments for the original and follow-up studies.

Natural-data transfer and model selection. The reserved-source comparison uses Traffic, Hospital, Saugeen River flow, and US births. Raw-history augmentation with KAHM features has 4.26% higher source-balanced scaled MSE than raw RBF and costs 2.30 times as much in the measured warm-library workflow. Raw ridge is more accurate than both. The favorable development mean is strongly influenced by FRED-MD, and it does not establish a KAHM-specific benefit. Adding geometric descriptors to metadata and a paid validation probe changes none of the 80 confirmation decisions. The post-hoc min–max scaling

Table 5: Scope and outcomes of the supporting studies. Counts describe the corresponding design, not independent forecasting domains. Natural source aggregates and post-hoc sensitivities are descriptive.

Study	Evaluation scope	Interpretation
Information and decoder capacity	Target interventions and 24 full repetitions per forecasting setting	Retaining local position can help; KAHM gains depend on the law and decoder.
Natural transfer and scaling	Eight source families; four reserved for confirmation; 208 configurations	Raw plus KAHM has higher reserved-source loss than raw RBF. A post-hoc scaling change preserves that ordering.
Additive correction	144 synthetic and 24 archive configurations	Some synthetic gains; two incomplete archive configurations prevent the planned full-roster comparison.
Proxy metric and donor residuals	48 tasks and 16 procedures	Corrections improve ridge, but incremental benefits of the learned metric and donor residuals are unresolved.
Direct reconstruction training	46 planned tasks; 43 feasible	Training helps KAHM in some settings; equally trained local controls remain competitive or better.
Online adaptation	Twelve streams, six methods, two capacities	Both capacity settings fail their accuracy criteria; extreme extrapolation persists after a solver correction.

analysis reuses these sources and simulation draws; it changes individual errors but does not reverse the principal mean-loss orderings.

Residual correction and learned geometry. An additive correction preserves the fitted raw baseline as a candidate, but validation can select a correction that worsens test error. The synthetic factorial resolves gains over raw RBF in five of twelve comparisons, while a simpler geometric correction has lower mean error in every Van der Pol and Duffing cell. On the archive roster, the two capacities of one Car Parts series produce eight failed geometric fits and four unavailable policy outputs. The planned full-roster geometric and policy losses are consequently undefined. A post-hoc comparison on 22 successful configurations gives additive KAHM only a 0.31% gain over raw RBF at 13.29 times its measured model-pipeline cost; ridge is more accurate and cheaper on that subset.

The donor-residual study completes all 48 tasks. Its learned-metric, out-of-fold KAHM procedure improves the ridge reference on the three synthetic laws and has 2.18% lower descriptive archive loss. However, the metric-versus-identity and donor-versus-in-sample intervals include zero. No adjusted comparison with an equally trained geometric control favors KAHM. These results support residual correction in the evaluated settings without identifying the proposed metric or KAHM reconstruction as the necessary source of that improvement.

Training and adaptation. Direct reconstruction training produces two favorable, six unfavorable, and ten unresolved primary contrasts. Neither favorable contrast is against an equally shape-trained non-KAHM local control. Three selected rideshare series have no eligible training-query windows, leaving the planned four-collection natural aggregate undefined; results on the seven available series form a separate descriptive comparison. A nine-task gate-training pilot uses a separate generator and budget and does not contribute to those intervals.

The online study meets its declared 99th-percentile service-time screen, but both capacity settings fail the recurring-stream and natural-data accuracy criteria. A corrected ridge solver removes computational exceptions while retaining an extreme forecast that a fresh solve also reproduces. Thus a nonvanishing reconstruction denominator and an accurate linear solve do not by themselves control feature extrapolation. The bounded, fixed-geometry decoder analyzed in the main experiment is a different procedure; its correction bound does not validate a repair of the online method.

8 Discussion and conclusion

The central results distinguish three properties of a reconstruction representation. Algebraic distinctness means that its coordinates permit functions unavailable to a specified comparison span. A bounded correction means that a fitted residual head cannot exceed a stated magnitude under the feature-norm assumption. Predictive usefulness means that the complete procedure achieves lower loss under a specified evaluation distribution. None of these properties implies the next one without additional assumptions or evidence.

The principal experiment supports a contribution from reconstruction departures beyond the shared locality and principal-coordinate features. It establishes the specified margins over independently selected common features and raw RBF, but not the complete practical-advantage criterion across the comparison family. RBF reconstruction achieves much of the same benefit. KAHM's point improvement over boosting is below the specified margin, and the selected non-KAHM procedure has lower aggregate loss. Failure to establish a margin does not establish equality of two methods. Likewise, the irregular-bank example demonstrates existence of distinct raw spans, not distinctness for every bank fitted in the experiment.

Local dimension affects several parts of the predictor simultaneously. It changes the common-coordinate basis, projected covariance and kernel, and reconstruction departure. In these experiments, selecting the cap improves the complete KAHM procedure relative to cap two, whereas the native default cap twenty increases aggregate error. The matched-bank decomposition shows that stronger common features account for the selected procedure's gain despite a smaller incremental benefit from departures. Full ambient local rank remains a valid native construction, but it is not an empirical guarantee of better forecasting. Since the history dimension is twelve, the observed cap-twenty procedure permits full ambient rank, not evidence about a binding twenty-dimensional cap on longer histories.

The correction bound provides a precise magnitude guarantee for the specified decoder. It does not establish robustness of the full history-to-feature map: signed normalization and the affine fallback may still be sensitive to the input. The baseline and targets are not uniformly bounded, so the theorem gives no uniform forecast-error bound. Its applicability to other bounded features also limits a claim of methodological novelty based on the bound alone.

The empirical scope is finite. Twenty-four independent repetitions quantify variability under four fixed simulation laws; they are not 24 independent application domains. The observation maps, finite burn-in, history length, horizons, model grids, and resource settings are part of the tested procedures. The simultaneous intervals use a paired-mean t approximation, and the 3% improvement and 5% noninferiority margins express research criteria rather than application-specific requirements. The supporting natural-data and adaptive studies impose further limits on extrapolation of the simulated gains. The experiments are not a comprehensive comparison with neural, pretrained, long-context multivariate, or modern adaptive forecasting methods.

Computation must also be interpreted at the level measured. Fitting costs include the candidates evaluated during selection. Prediction time, serialized state size, owned array storage, and whole-process peak memory measure different resources and cannot be substituted for one another. The dimension-selected procedure pays for every evaluated cap; charging only its winning bank would understate its selection cost. The measurements characterize the supplied implementations on the recorded environment; they do not establish minimum algorithmic costs or satisfy an unspecified deployment budget.

KAHM reconstruction can therefore supply useful features to a finite forecasting decoder, with a value that depends on geometry, feature parameterization, regularization, and target structure. The contribution of the present study is to characterize these dependencies and evaluate them against matched alternatives. A general forecasting advantage specific to KAHM remains unestablished by the reported evidence.

Reproducibility and research transparency

The accompanying computational materials retain the protocol, source freeze, fitted states, predictions, input and target arrays, selection records, and analysis for all 192 task suites under `extensions/dimension_forecasting`. The analysis program verifies the source and stage inventories before reconstructing the 27 numerical tables and six primary contrasts. Every cap, all ten procedures, and all declared conditions are

retained regardless of predictive performance. The complete prediction inventory contains 31,104 arrays and 155,271,168 forecast values, including fixed-cap, selected, matched-bank, calibrated, and unit-calibration views. The ten calibrated primary procedures alone account for 1,920 arrays and 9,584,640 forecast values.

An independent audit regenerates episodes from the pinned generators and independently checks seeds, noise, windows, and target construction, the selection and calibration calculations, all 18,816 stored local bases, and all 261,450 rows of the released analysis tables. Numerical parity checks against the original full comparison suite use manufactured inputs before the source freeze. These are computational consistency checks, not an independent scientific replication. Reconstructing scores from saved forecasts, replaying fitted states, and refitting candidates remain distinct operations. The archived arrays define the reported scores; reproducing floating-point states requires compatible numerical libraries and arithmetic.

The standalone manuscript source includes its generated displays and compiles without the numerical archives. Its asset builder reads the sealed study tables and records their checksums. The companion *Supplementary Studies* gives the complete methods and results for the supporting investigations in Section 7. The complete principal fitting specification is in Appendix B. The prospective specification was local, informed by earlier studies, and not publicly preregistered.

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A Reconstruction algebra and bounded residual decoding

This appendix analyzes fixed reference maps and their use by a specified decoder. It establishes normalized-kernel identities, conditions for equality or inequality of feature spans, and a bound on fitted residual corrections. None of these statements assumes a particular time-series distribution or guarantees a forecasting improvement. Unless another subscript is displayed, vector norms are Euclidean norms. The numerical examples can be evaluated with `extensions/reconstruction_mechanism/algebra.py`; their reference banks are constructed independently of forecasting outcomes.

A.1 Normalized kernel identities

Let $q_1, \ldots, q_M \in \mathbb{R}^d$ be reference histories, and let $Q \in \mathbb{R}^{M \times d}$ contain their transposes as rows. Let $P \in \mathbb{R}^{d \times r}$ have orthonormal columns, with $1 \le r \le d$, and let $W \in \mathbb{R}^{r \times r}$ be a symmetric positive-definite precision matrix for the projected coordinates. For a column-vector query $z \in \mathbb{R}^d$, define the Gaussian kernel vector $k(z) \in \mathbb{R}^M$ by

$$k_i(z) = \exp\left[-\frac{(P^\top(z - q_i))^\top W (P^\top(z - q_i))}{2r}\right], \qquad i = 1, \ldots, M.$$

Let $K \in \mathbb{R}^{M \times M}$ be the reference Gram matrix, with $K_{ij} = k_i(q_j)$. For $\lambda > 0$, set $C = (K + \lambda I_M)^{-1}$, where I_M is the $M \times M$ identity matrix, and let $\mathbf{1}_M$ be the length-M vector of ones. The matrix K is symmetric positive semidefinite, so C exists and is symmetric positive definite. The KAHM and matched positive-kernel reconstructions are

$$A_K(z) = \frac{Q^\top C k(z)}{\mathbf{1}_M^\top C k(z)}, \qquad A_R(z) = \frac{Q^\top k(z)}{\mathbf{1}_M^\top k(z)}. \tag{9}$$

The first expression is defined wherever its signed denominator is nonzero. The second denominator is positive for every finite query. The two maps share the references and projected Gaussian kernel. Multiplying $k(z)$ by a common positive scalar cancels in either ratio, which justifies subtracting the largest log-kernel value before query evaluation.

Target-transfer identity. For a query satisfying $s(z) = \mathbf{1}_M^\top C k(z) \neq 0$, define the normalized weight vector $\omega(z) = Ck(z)/s(z)$. Its entries sum to one. For any reference target vector $y \in \mathbb{R}^M$,

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1} \mathbf{1}_M}.$$

Thus target transfer is the ratio of two dual kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept: one fitted to y, and one fitted to the constant target $\mathbf{1}_M$. For an affine scalar function $f(q) = q^\top b + c$, with $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, let $f(Q) = Qb + c\mathbf{1}_M$ be its reference values. Then

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

Proof of Proposition 1. The definition gives $\mathbf{1}_M^\top \omega(z) = s(z)/s(z) = 1$. Because $C = C^\top$, $\omega(z)^\top y = k(z)^\top C y / s(z)$ and $s(z) = k(z)^\top C \mathbf{1}_M$, proving the ratio identity. For the affine function,

$$\omega(z)^\top f(Q) = \omega(z)^\top Q b + c\omega(z)^\top \mathbf{1}_M = A_K(z)^\top b + c.$$

Subtracting $z^\top b + c$ proves the final identity. □

This identifies direct target transfer as a normalized kernel-ridge construction. The ratio generally differs from the unnormalized kernel-ridge predictor because its denominator depends on the query. Similarly, $A_K(z)^\top b$ is normalized kernel ridge applied to the restricted reference target Qb. For any reconstruction A, the identity $A(z) + \{z - A(z)\} = z$ also implies a linear dependence among the raw input, reconstruction, and residual when all three are included as unrestricted linear features. These identities constrain claims about the resulting function class while allowing useful parameterizations for a restricted or regularized decoder.

A.2 When reconstruction departures have the same span

Let $\mu \in \mathbb{R}^d$ be the reference center and define the affine reconstruction $A_A(z) = \mu + PP^\top(z - \mu)$. Fix a common scalar locality gate $g(z)$, and define column-vector departures

$$d_K(z) = g(z)\{A_K(z) - A_A(z)\}, \qquad d_R(z) = g(z)\{A_R(z) - A_A(z)\}.$$

A feature span means the set of scalar functions obtained by finite linear combinations of its coordinate functions. Let $\mathcal{S}_0$ be the span of the gate $g(z)$ and the coordinates of $g(z)P^\top(z - \mu)$. It contains every coordinate of $g(z)A_A(z)$.

Equal augmented feature spans. Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ and C leaves the augmented reference-coordinate space $\mathcal{V} = \mathrm{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\} \subseteq \mathbb{R}^M$ invariant. Here $Q_{:j}$ denotes column j of Q. Then there exist $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad a = (\kappa+\lambda)^{-1} > 0.$$

For these coefficients,

$$A_K = \frac{B^\top A_R + c}{a}, \qquad d_K = \frac{B^\top}{a} d_R + \left(\frac{B^\top}{a} - I_d\right) g A_A + \frac{gc}{a}.$$

The span obtained by adjoining the coordinates of d_K to $\mathcal{S}_0$ is the same as that obtained by adjoining the coordinates of d_R.

Proof of Proposition 2. The constant-row-sum assumption implies $(K+\lambda I_M)\mathbf{1}_M = (\kappa+\lambda)\mathbf{1}_M$, so $C\mathbf{1}_M = a\mathbf{1}_M$. Since K is positive semidefinite, $\kappa \geq 0$ and hence $a > 0$. Invariance of $\mathcal{V}$ means each column of CQ can be expressed using Q and $\mathbf{1}_M$, which gives B and c. Using symmetry of C in Equation (9),

$$A_K(z) = \frac{(CQ)^\top k(z)}{(C\mathbf{1}_M)^\top k(z)} = \frac{B^\top Q^\top k(z) + c\mathbf{1}_M^\top k(z)}{a\mathbf{1}_M^\top k(z)} = \frac{B^\top A_R(z) + c}{a}.$$

Subtract A_A and multiply by g to obtain the displayed departure identity. Since $gA_A = g\mu + P\{gP^\top(z-\mu)\}$, all its coordinates belong to $\mathcal{S}_0$. Thus the KAHM-augmented span is contained in the RBF-augmented span.

For the reverse inclusion, the restriction of the invertible matrix C to the invariant finite-dimensional space $\mathcal{V}$ is injective and therefore surjective onto $\mathcal{V}$. Its images $C\mathbf{1}_M = a\mathbf{1}_M$ and the columns of CQ consequently span $\mathcal{V}$. Hence there exist $D \in \mathbb{R}^{d\times d}$ and $e \in \mathbb{R}^d$ such that $Q = (CQ)D + \mathbf{1}_M e^\top$. Multiplying its transpose by $k(z)/(\mathbf{1}_M^\top k(z))$ gives $A_R(z) = aD^\top A_K(z) + e$. After multiplication by g and subtraction of gA_A, this expresses each RBF departure coordinate using KAHM departure coordinates and $\mathcal{S}_0$. The spans therefore agree, including when the original reference-coordinate columns are redundant. □

For example, take scalar references $q = (-1, 1)^\top$, $P = 1$, $g = 1$, and $A_A(z) = z$. If the off-diagonal Gram entry is ρ, then $K\mathbf{1}_2 = (1+\rho)\mathbf{1}_2$ and $Kq = (1-\rho)q$. Therefore

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1+\rho+\lambda}{1-\rho+\lambda}. \tag{10}$$

The coefficient c_ρ is positive, so $\mathrm{span}(1, z, d_K) = \mathrm{span}(1, z, d_R)$. With the sample-variance precision $1/2$, $\rho = e^{-1}$. The regularization iteration used by the implementation yields $\lambda = 2.6519665429866075$ and $c_\rho = 1.224037566462905$. The maximum numerical identity error on the constructed query grid is below 6×10^{-16}.

These are equalities between untransformed feature spans. An isotropic ridge penalty is not invariant under a general nonorthogonal change of coordinates. Applying separate nonlinear radial transformations can also remove the displayed linear relation. A forecasting comparison therefore evaluates the complete reconstruction, feature transformation, and fitting procedure, even when two untransformed feature sets have the same span.

A.3 An explicit distinct reconstruction basis

Irregular scalar references. Let $q_1 < q_2 < q_3$ with $q_1 + q_3 \neq 2q_2$, and define $k_i(z) = \exp\{-t(z-q_i)^2/2\}$ for precision $t > 0$. Let $K_{ij} = \exp\{-t(q_i - q_j)^2/2\}$, $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$, and suppose every entry of $v = C\mathbf{1}_3$ is positive. For the reconstructions in Equation (9), with $Q = q = (q_1, q_2, q_3)^\top$, gate $g = 1$, and $A_A(z) = z$,

$$A_K \notin \mathrm{span}(1, z, A_R), \qquad d_K \notin \mathrm{span}(1, z, d_R).$$

These are statements about scalar functions on the whole real line.

Proof of Proposition 3. Set $u = Cq$. Since C is symmetric, $A_K(z) = (k(z)^\top u)/(k(z)^\top v)$, and positivity of v makes its denominator positive for every z. The six sums $q_i + q_j$ with $i \le j$ are distinct: strict ordering of the three references orders all the sums except possibly $q_1 + q_3$ and $2q_2$, and their equality is excluded by assumption. Products of Gaussian translates satisfy

$$k_i(z)k_j(z) = e^{-tz^2} e^{-t(q_i^2+q_j^2)/2} e^{t(q_i+q_j)z}.$$

After removal of the common positive factor e^{-tz^2}, these products are nonzero multiples of exponentials with six distinct real exponents. Such exponentials are linearly independent. Indeed, differentiating a vanishing linear combination zero through fifth order at $z = 0$ gives a Vandermonde system whose determinant is the product of the nonzero pairwise exponent differences.

Suppose first that $A_K = cA_R + d$ for scalars c, d. Define $w = u - dv$. Multiplication by the two positive denominators gives

$$\left(\sum_i k_i w_i\right)\left(\sum_j k_j\right) = c\left(\sum_i k_i q_i\right)\left(\sum_j k_j v_j\right).$$

Independence of the six products implies equality of their coefficients. The diagonal coefficients give $w_i = cq_i v_i$, and the off-diagonal coefficients give $w_i + w_j = c(q_i v_j + q_j v_i)$ for $i < j$. If $c = 0$, all w_i vanish, so $Cq = dC\mathbf{1}_3$. Invertibility of C would imply $q = d\mathbf{1}_3$, contrary to the distinct references. If $c \neq 0$, substituting the diagonal equations into the off-diagonal ones yields $(q_i - q_j)(v_i - v_j) = 0$. Thus $v = a\mathbf{1}_3$ for some $a > 0$, and $(K + \lambda I_3)v = \mathbf{1}_3$ implies that K has constant row sums. This is impossible: its middle-row sum exceeds the first-row sum by $K_{23} - K_{13} > 0$ and the last-row sum by $K_{12} - K_{13} > 0$, since the Gaussian decreases strictly with positive distance. This rules out every affine relation $A_K = cA_R + d$.

Both reconstructions are bounded functions of z. The RBF reconstruction is a convex combination of q_1, q_2, q_3. Also,

$$A_K(z) = \frac{\sum_i k_i(z) v_i \, (u_i/v_i)}{\sum_i k_i(z) v_i}$$

is a convex combination of the three finite numbers u_i/v_i. If $A_K(z) = cA_R(z) + d + bz$ held on the real line, boundedness of both reconstructions would force $b = 0$. The affine relation already ruled out would then follow. This proves the first span statement. Finally, $d_K = A_K - z$ and $d_R = A_R - z$ imply $\mathrm{span}(1, z, d_R) = \mathrm{span}(1, z, A_R)$. Membership of d_K in this span would imply membership of A_K, a contradiction. □

A concrete example uses $q = (-1, 0, 2)^\top$, sample-variance precision $t = 3/7$, and $\lambda = 4.498274743339644$ from the specified regularization iteration. Positivity of $C\mathbf{1}_3$ can be verified without examining a query grid. Write $K = I_3 + E$ and $a = 1 + \lambda$. For a matrix, let $\|\cdot\|_\infty$ denote the maximum absolute row sum; for a vector, it denotes the largest absolute entry. If $a > 2\|E\|_\infty$, then the Neumann series for $aC = (I_3 + E/a)^{-1}$ gives

$$\|aC\mathbf{1}_3 - \mathbf{1}_3\|_\infty \le \sum_{j=1}^{\infty} (\|E\|_\infty/a)^j = \frac{\|E\|_\infty/a}{1 - \|E\|_\infty/a} < 1.$$

Every entry of $aC\mathbf{1}_3$ is then strictly positive. In the example, $\|E\|_\infty = e^{-3/14} + e^{-6/7} = 1.2314905926823394$, so the condition holds. On 81 constructed queries in $[-3, 4]$, the maximum absolute residual after least-squares projection of the sampled d_K values onto the sampled $(1, z, d_R)$ columns is 0.05355670997078987. This numerical illustration supplements the analytic existence result. It does not establish distinctness for every higher-dimensional bank used in the forecasting experiments.

A.4 A bounded correction with a free intercept

For a feature block $v \in \mathbb{R}^{p_b}$ of positive width p_b, define

$$T_{R_b}(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R_b^2}}, \qquad R_b = \sqrt{p_b}. \tag{11}$$

Its norm is at most R_b. To see its sensitivity to its own input, set $a = 1 + \|v\|_2^2/R_b^2$. Its Jacobian is $a^{-1/2}I - a^{-3/2}vv^\top/R_b^2$. For $v \neq 0$, the eigenvalue in the direction of v is $a^{-3/2}$, and each orthogonal direction has eigenvalue $a^{-1/2}$; at zero, the Jacobian is the identity. All its singular values are therefore at most one. Integrating the derivative along the segment between any two inputs proves $\|T_{R_b}(v) - T_{R_b}(v')\|_2 \leq \|v - v'\|_2$. Thus the radial transformation is nonexpansive.

Concatenating separately transformed blocks gives a feature vector $\psi(z) \in \mathbb{R}^p$, where $p = \sum_b p_b$, satisfying $\|\psi(z)\|_2 \leq R = (\sum_b R_b^2)^{1/2}$. This norm bound holds wherever the preceding feature construction is defined. Nonexpansiveness of the radial transformation does not by itself give a Lipschitz constant for the full history-to-feature map, which includes the signed reconstruction denominator and its fallback rule.

Centered ridge correction. For $n \geq 1$, let $e_i \in \mathbb{R}$ be residual targets and $\psi_i \in \mathbb{R}^p$ be training feature vectors. Fit $b \in \mathbb{R}$ and $w \in \mathbb{R}^p$ by minimizing

$$\frac{1}{n}\sum_{i=1}^{n}(e_i - b - \psi_i^\top w)^2 + \alpha\|w\|_2^2, \qquad \alpha > 0,$$

with no penalty on b. Define the training means $\bar{e} = n^{-1}\sum_i e_i$ and $\bar{\psi} = n^{-1}\sum_i \psi_i$, the residual vector $e = (e_1, \ldots, e_n)^\top$, its centered version $e_c = e - \bar{e}\mathbf{1}_n$, and its centered root mean square $\sigma_e = \|e_c\|_2/\sqrt{n}$. Suppose query features satisfy $\|\psi(z)\|_2 \leq R$. Then, for any scalar γ,

$$\|w\|_2 \leq \frac{\sigma_e}{2\sqrt{\alpha}}, \qquad |\gamma q(z)| \leq |\gamma|\left[|\bar{e}| + (R + \|\bar{\psi}\|_2)\frac{\sigma_e}{2\sqrt{\alpha}}\right], \quad q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w. \tag{12}$$

Proof of Proposition 4. For fixed w, differentiating with respect to the intercept gives $b = \bar{e} - \bar{\psi}^\top w$. Let $\Psi_c \in \mathbb{R}^{n\times p}$ have row i equal to $(\psi_i - \bar{\psi})^\top$. Substituting the fitted intercept reduces the objective to $n^{-1}\|e_c - \Psi_c w\|_2^2 + \alpha\|w\|_2^2$. Its unique minimizer is

$$w = (\Psi_c^\top\Psi_c + n\alpha I_p)^{-1}\Psi_c^\top e_c.$$

For a singular value $s \geq 0$ of Ψ_c, the corresponding singular value of the coefficient operator is $s/(s^2 + n\alpha)$. The inequality $(s - \sqrt{n\alpha})^2 \geq 0$ implies

$$\frac{s}{s^2 + n\alpha} \leq \frac{1}{2\sqrt{n\alpha}}.$$

The operator norm is therefore at most $1/(2\sqrt{n\alpha})$, giving $\|w\|_2 \leq \|e_c\|_2/(2\sqrt{n\alpha}) = \sigma_e/(2\sqrt{\alpha})$. Finally, the fitted intercept gives $q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w$. Cauchy–Schwarz and the triangle inequality yield

$$|q(z)| \leq |\bar{e}| + (\|\psi(z)\|_2 + \|\bar{\psi}\|_2)\|w\|_2 \leq |\bar{e}| + (R + \|\bar{\psi}\|_2)\frac{\sigma_e}{2\sqrt{\alpha}}.$$

Multiplication by $|\gamma|$ proves the correction bound. □

The mean-squared-error objective corresponds to a summed-squared-error objective with penalty $n\alpha$. Keeping the numerical value α unchanged when switching to a summed loss would define a different estimator. For multiple forecast horizons, the bound applies separately to each fitted residual vector and each selected pair (α_h, γ_h). This is a general property of ridge regression on bounded features, including the KAHM features considered here. It bounds the additive correction but does not bound the retained raw baseline or total forecast error. The experiment checks the feature, coefficient, and correction inequalities on every saved task and reports forecast errors separately.

B Complete reconstruction-departure evaluation

This appendix specifies the observation laws, fitting procedure, complete comparison set, stability checks, and resource measurements for Section 6. The reproduction materials contain the implementation, saved predictions and model states, and machine-readable analyses under `extensions/dimension_forecasting`. All tables and figures in this appendix are generated from those analyses.

B.1 Observation laws and data roles

The linear negative control is

$$y_t = 0.65y_{t-1} - 0.15y_{t-2} + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.1^2),$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. The process starts with two zeros; 256 stored values are discarded, and 72 values beginning at index 256 are retained. The nonlinear autoregression is

$$y_t = \tanh\{1.25y_{t-1} - 0.65y_{t-2} + 0.25\sin(3y_{t-1}) + u_t\}, \qquad u_t \overset{\text{iid}}{\sim} U[-0.1, 0.1],$$

where $U[a, b]$ denotes the uniform distribution on $[a, b]$. Its two initial values are independent draws from $U[-0.8, 0.8]$, and 80 stored values are discarded before retaining an episode.

The Van der Pol dynamics are $\dot{x} = v$ and $\dot{v} = (1 - x^2)v - x$. The Duffing dynamics are $\dot{x} = v$ and $\dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi)$. Here x is position, v is velocity, a dot denotes differentiation with respect to continuous time t, and the Duffing phase φ is uniform on $[0, 2\pi]$. Initial position and velocity are independent draws from $U[-2, 2]$. The ordinary differential equations are integrated in float64 arithmetic using the classical fourth-order Runge–Kutta method with step 0.025, observation interval 0.1, and a discarded initial time interval of length ten. The scalar observation is $0.95\tanh(x) + \xi$, with independent sensor noise $\xi \sim U[-0.02, 0.02]$. The resulting numerical integration and observation map define the tested simulation law; the observations are not exact continuous-time trajectories.

For each law and repetition, the episode sets A, B, C, and D are generated once. Additional noise has the form $\eta s_A Z$, where $\eta \in \{0, 0.1\}$, $Z \sim \mathcal{N}(0, 1)$ independently at each stored observation, and s_A is the empirical standard deviation, with divisor equal to the number of observations, of all A episode values before this addition. The underlying episodes and Z arrays are paired across the two noise levels. Added noise affects both histories and observed future targets. At $\eta = 0$, the oscillator sensor noise is still present. The B/C/D linear observations are Gaussian conditional on the A-estimated noise amplitude, while their marginal law across repetitions is a variance mixture. Finite burn-in does not imply exact stationarity, and the added Gaussian noise means that the observed responses are not uniformly bounded.

The master seed is 2026110103 and repetitions are indexed by $r = 0, \ldots, 23$. A uint32 stream seed is the first output of NumPy's `SeedSequence` applied to an integer tuple. Law indices follow linear AR, nonlinear AR, Van der Pol, and Duffing; set indices are A/B/C/D = 0/1/2/3. Base, noise, and retained-row streams use $(2026110103, r, \text{law}, \text{set}, \text{purpose})$, with purposes 1001, 1002, and 1003. Fitting uses $(2026110103, r, \text{law}, 1004)$, and the repetition identifier uses $(2026110103, r, 1000)$. The 1,272 intended-distinct stream identifiers are checked for collisions and for overlap with the 2,173 distinct identifiers recorded for earlier principal and dimension studies. Base episodes and standardized noise arrays for different sets use independent streams. The additional-noise amplitude is estimated on A and shared across sets. Fitting seeds, row choices, references, and noise draws are paired across caps and noise levels. Using zero-based indices, the 13 origins $11, 15, \ldots, 59$ each have a length-12 history ending at the origin and targets at offsets 1, 4, and 12. A/B/C/D contain 64/32/32/128 episodes. The first three sets retain 512/256/256 histories; D retains all 1,664 histories. This gives 958,464 scored target values per method over the full experiment. Inference nevertheless uses 24 independent repetition aggregates.

A fits every scaler, reference bank, kernel factor, baseline, and decoder. B selects penalties and raw-model settings separately by horizon, then one dimension cap across horizons for each geometric procedure. Its cap objective uses the uncalibrated predictions in Equation (7); all five cap choices are saved before C is loaded. C selects each candidate's mixing coefficients and then one complete non-KAHM procedure within each comparison context. All fitted states and the complete task roster are sealed before D episodes are generated. D histories are supplied to fixed predictors; their targets remain separate until every prediction is sealed. A later origin can legitimately contain observations that were targets for an earlier origin in the same episode, but it causes no fitting or cross-query adaptation. All uncertainty calculations use independent repetitions, not the overlapping origins.

B.2 Reference geometry and numerical evaluation

The geometric bank first standardizes each history coordinate using its A mean and standard deviation. A standard deviation at most 10^{-12} is replaced by one. At most 512 A rows enter the bank fit. Separate k-means partitions request 2, 4, and 8 groups, using five initializations and at most 150 iterations. If there are too few distinct input rows, the requested group counts are reduced to the available number. An unresolved empty-group configuration is retried with halved group counts. Within each group, at most 24 reference rows are sampled without replacement. The references are stored in float32. The native PCA converts its input to float64 and uses the centered sample covariance. Native candidate cap $c \in \{1, 2, 4, 8, 12, 20\}$ initializes the requested rank at $\min\{c, d, n_j - 1\}$. Eigenvalues must exceed double-precision machine epsilon. The requested rank is decremented and the native PCA recomputed if the smallest range of a projected reference coordinate is below 10^{-3}, stopping when the range condition holds or requested rank one is reached. A rank-one model may therefore remain below that threshold. The explicit fixed-cap-two comparison instead initializes $\min\{2, d-1, n_j - 1\}$ and uses the same subsequent numerical rules. The actual stored rank is the number of columns of P_j, not the supplied cap. The forecasting wrapper preserves its explicit zero-rank affine fallback for singleton or constant references; this is not claimed as successful native PCA. All cap requests, initial ranks, stored ranks, reference counts, and reduction or fallback reasons are recorded.

For the row-stored references Q_j, define $\mu_j = n_j^{-1} \sum_{i=1}^{n_j} q_{ji}$ and

$$s_j = \max\left\{ \frac{1}{n_j} \sum_{i=1}^{n_j} \|q_{ji} - \mu_j\|_2^2, 10^{-6} \right\}.$$

This is the scale used in the common locality weights. The projected sample covariance is

$$S_j = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} P_j^\top (q_{ji} - \mu_j)(q_{ji} - \mu_j)^\top P_j.$$

Its inverse defines the shared Gaussian kernel in Section 3. Kernels and references are identical between KAHM and positive-RBF reconstruction within a bank. Independent cap choices can select different projected kernels and bases.

The KAHM regularizer is fitted from reference reconstruction, without forecast targets. Suppress the local index j, let $Q \in \mathbb{R}^{n \times d}$ contain the local references, and define the reference energy $\nu = \|Q\|_F^2/(nd)$, where $\|\cdot\|_F$ is the square root of the sum of squared matrix entries. Initialize $\lambda^{(0)} = 2.5\nu$. At iteration a, compute

$$C^{(a)} = (K + \lambda^{(a)} I_n)^{-1}, \qquad \lambda^{(a+1)} = 2\nu + \frac{\|Q - KC^{(a)}Q\|_F^2}{nd}.$$

The iteration stops when $|\lambda^{(a+1)} - \lambda^{(a)}| \le 10^{-3}$ or after 100 inverse computations. The reconstruction uses the last computed $C^{(a)}$ and its corresponding $\lambda^{(a)}$, rather than the subsequent proposal $\lambda^{(a+1)}$. An unexpected factorization failure or invalid regularizer stops the experiment; it is not converted into a successful candidate. This rule determines a reference-based regularizer; it does not optimize forecasting loss.

For each query, the maximum log-kernel entry is subtracted before exponentiation. The resulting common positive rescaling leaves both normalized reconstruction ratios unchanged. Signed cancellation is checked using ρ_j in Equation (3). The affine fallback makes the corresponding raw departure zero if the ratio cannot be evaluated reliably. The experimental results below report how often this occurs.

B.3 Decoders, comparison grids, and descriptive losses

The common and departure feature columns are standardized separately using A, with each standard deviation replaced by its maximum with 10^{-6}. The blocks are then radially bounded as specified in Equation (5), with no subsequent scaling. The two unbounded-departure variants retain the bounded common block and omit only the departure's radial transformation.

Every ridge head uses mean squared error plus $\alpha\|w\|_2^2$ and an unpenalized intercept, with $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. Equivalently, the unnormalized sum-of-squares objective uses penalty $n\alpha$ for n

fitting rows. Raw-history ridge and RBF methods standardize their input coordinates on A with a 10^{-6} scale floor. The raw RBF method concatenates standardized histories with at most 64 Nyström features, then standardizes the concatenated columns using A. The squared Gaussian bandwidth equals the median of all strictly positive squared landmark distances multiplied by 0.5, 1, or 2, with no bandwidth floor. When there are no strictly positive landmark distances, the median used by this rule is set to one. B selects this factor and the ridge penalty separately for each horizon. For a landmark Gram matrix whose largest eigenvalue is $\zeta_{\max}$, the Nyström transform retains eigenvalues above $\max\{10^{-12}, 10^{-10}\zeta_{\max}\}$. Let K_{MM} be the symmetrized landmark Gram matrix. Let the columns of V be its retained orthonormal eigenvectors, and let the diagonal matrix Λ contain their positive eigenvalues. If $k_M(u)$ is the vector of Gaussian similarities between a standardized query u and the M landmarks, its Nyström coordinates are $\Lambda^{-1/2}V^\top k_M(u)$. The Gaussian is $\exp\{-\|u-u'\|_2^2/(2\ell^2)\}$, with squared bandwidth ℓ^2 chosen by the median-distance rule above. These coordinates are concatenated with the standardized raw history before column scaling and ridge fitting.

Histogram gradient boosting uses 100 iterations, L2 regularization one, and no early stopping. B selects learning rate 0.05 or 0.1 and maximum leaf count 7 or 15 separately by horizon. Persistence predicts the latest observed history value at every horizon. Every non-ridge candidate is then blended with ridge using a C-selected coefficient from $\{0, 1/4, 1/2, 1\}$. Ties are resolved by the recorded candidate order. The non-KAHM policy chooses in this order among ridge, raw RBF, boosting, persistence, the common basis, and the two RBF-departure variants; both KAHM variants are excluded. Table 6 reports all methods and conditions.

Table 6: All ten procedures, every declared law and both added-noise levels. Entries are mean horizon-averaged A-variance-normalized D MSE over 24 repetitions; lower is better. The first six columns enter the nonlinear aggregate with equal weight. The final two form the separate linear control. These conditionwise summaries are descriptive. "Unbounded" omits only the departure radial bound. Persistence and raw nonlinear controls include their C-selected blend with ridge.

Procedure	Nonlinear AR		Van der Pol		Duffing		Linear AR	
	$\eta = 0$	0.1	$\eta = 0$	0.1	$\eta = 0$	0.1	$\eta = 0$	0.1
Raw ridge	0.02200	0.03451	0.07470	0.13584	0.01955	0.06873	0.90079	0.90253
Raw RBF	0.01996	0.03243	0.01326	0.06486	0.01296	0.04511	0.90383	0.90556
Boosted trees	0.02038	0.03231	0.00380	0.05863	0.01351	0.04908	0.90422	0.90473
Persistence	0.02205	0.03451	0.07470	0.13588	0.01975	0.06955	0.90140	0.90382
Common features	0.01928	0.03115	0.02126	0.05902	0.01406	0.04587	0.90139	0.90319
Bounded KAHM departure	0.01928	0.03109	0.00935	0.05735	0.01147	0.04494	0.90230	0.90454
Bounded RBF departure	0.01930	0.03119	0.01064	0.05640	0.01289	0.04431	0.90198	0.90366
Unbounded KAHM departure	0.01949	0.03149	0.01013	0.05929	0.01097	0.04511	0.90237	0.90459
Unbounded RBF departure	0.01937	0.03132	0.01104	0.05861	0.01157	0.04471	0.90244	0.90385
Selected non-KAHM	0.01923	0.03110	0.00380	0.05615	0.01175	0.04463	0.90600	0.90700

The machine-readable files `horizon_losses.csv` and `condition_horizon_summary.csv` retain every method, law, noise level and horizon, including raw MSE and the A-only loss scale. They are not pooled with historical losses that use different normalizations. The `primary_repetition_contrasts.csv` file contains all 24 margin-gap values for each primary contrast, allowing the six lower bounds to be recomputed without importing any forecasting model.

B.4 Dimension selection and realized geometry

Table 8 reports all five geometric procedures' dimension choices. Native caps twelve and twenty have identical arrays and forecasts at ambient dimension twelve, and exact tie resolution selects twelve. Native cap two and the fixed-cap-two policy are also identical. These identities are checked on bases, features, fitted heads, B choices, C calibration and final predictions; the policy metadata remain distinct. Raw ridge, raw RBF, boosting, and persistence are fitted once per task and shared across the three contexts. The cap-dependent heads and seven banks are fitted separately without outcome-dependent deduplication.

There are 192 banks and 2,688 local models per cap rule. Local reference counts range from 2 to 24; 2,670 local reference sets per rule contain 24 rows. Table 9 reports stored ranks and feature widths. For native cap twenty, 2,685 local models have rank twelve and the remaining three have two references and rank one. All banks have 14 local models and a stacked principal-basis rank of twelve. No additional positive-eigenvalue or

Table 7: Nonlinear horizon-specific SMSE, averaged equally over the six law–noise conditions and 24 repetitions. Except for the first two fixed KAHM rows, procedures use their independently selected caps where applicable. Every non-ridge forecast includes its C calibration. These descriptive summaries retain the original A-only variance scales.

Procedure	Horizon 1	Horizon 4	Horizon 12
KAHM, fixed cap 2	0.011654	0.021471	0.056529
KAHM, native cap 20	0.011207	0.019108	0.066862
Raw ridge	0.013041	0.036954	0.127674
Raw RBF	0.011248	0.019724	0.063314
Boosted trees	0.011742	0.020626	0.056490
Persistence	0.013056	0.036971	0.128193
Common features	0.011149	0.018970	0.065207
Bounded KAHM departure	0.011201	0.019491	0.056043
Bounded RBF departure	0.011204	0.019516	0.056647
Unbounded KAHM departure	0.011378	0.020254	0.056611
Unbounded RBF departure	0.011226	0.019790	0.057292
Selected non-KAHM	0.011281	0.019431	0.052623

Table 8: B-selected cap counts. Each procedure contributes 144 nonlinear and 48 linear tasks; it selects one cap across all three horizons before C calibration. Exact cap-twelve/cap-twenty ties explain the zero final column. The common and RBF rows use their own choices, not KAHM's choices.

Procedure	Panel	1	2	4	8	12	20
Common features	Nonlinear	13	10	68	40	13	0
Common features	Linear	35	9	3	0	1	0
Bounded KAHM departure	Nonlinear	15	22	68	31	8	0
Bounded KAHM departure	Linear	4	14	18	9	3	0
Bounded RBF departure	Nonlinear	26	29	58	26	5	0
Bounded RBF departure	Linear	14	24	3	5	2	0
Unbounded KAHM departure	Nonlinear	17	27	67	25	8	0
Unbounded KAHM departure	Linear	2	5	27	11	3	0
Unbounded RBF departure	Nonlinear	19	39	59	20	7	0
Unbounded RBF departure	Linear	9	26	10	2	1	0

projected-range reduction occurs in this sample. The observed ranks describe these banks, not a guarantee for arbitrary datasets.

Table 9: Realized geometry over the complete task roster. Rank statistics refer to the 2,688 local models in each row; "Rank 12" counts full-ambient local bases. Width ranges refer to 192 complete banks. The full departure readout adds the common and departure widths. Stacked-basis rank is twelve for every bank, even when its individual local ranks are small.

Bank rule	Rank range	Mean rank	Rank 12	Common width	Departure width
Fixed cap 2	1–2	1.999	0	40–42	168
Native cap 1	1–1	1.000	0	28–28	168
Native cap 2	1–2	1.999	0	40–42	168
Native cap 4	1–4	3.997	0	64–70	168
Native cap 8	1–8	7.992	0	112–126	168
Native cap 12	1–12	11.988	2685	160–182	168
Native cap 20	1–12	11.988	2685	160–182	168

The common and departure contributions in Section 6 use the KAHM-selected bank, not the independently selected common procedure. The matched common mean loss is 0.032702, compared with 0.031775 for independently selected common features. Table 10 reports the exact mean loss decomposition relative to fixed cap two. For native cap twenty, the incremental departure benefit is only -0.000653, compared with -0.004967 at cap two. A change in complete KAHM accuracy must therefore be distinguished from a change in the incremental value of its reconstruction departure.

Table 10: Nonlinear loss decomposition relative to the fixed-cap-two procedure. Total change equals common change plus increment change. The final column gives the new KAHM-minus-common loss on the same bank. Negative values in the first two columns denote improvement; a positive increment change denotes a weaker departure benefit. These are descriptive identities between calibrated predictor losses.

Bank rule	Total change	Common change	Increment change	$L_K - L_0$
KAHM-selected bank	-0.000973	-0.002150	0.001177	-0.003790
Native cap 20	0.002507	-0.001806	0.004313	-0.000653

B.5 Stability, numerical safeguards and correction activation

The bounded-feature checks cover 192 tasks, seven banks, all four data stages, and each of the three bounded procedures. Coefficient checks cover all five geometric heads and three horizons for each task and bank; global correction checks apply to the three bounded heads. The independent analyzer checks the recorded fitted-state bound formulas and recomputes actual D corrections as the saved forecast minus the saved raw ridge forecast. It does not claim to reconstruct the fitted coefficient matrices from summary statistics alone. Table 11 reports maxima over all seven banks and the complete roster; maxima in different columns need not come from the same task or horizon.

Table 11: Observed ratios to the deterministic bounded-feature and ridge-readout limits. Feature ratios concern each procedure's complete feature vector and its actual radius, including both blocks for the two appended procedures. A zero correction with a zero bound contributes ratio zero. "Violations" sums failed feature, coefficient and correction checks at the fixed numerical tolerances. These are implementation checks, not confidence intervals for forecast risk.

Procedure	Feature ratio	Coefficient ratio	Correction ratio	Violations
Common features	0.9935	0.7154	0.2824	0
Bounded KAHM departure	0.9983	0.6735	0.2207	0
Bounded RBF departure	0.9974	0.6738	0.2347	0

The maximum taskwise ratio of KAHM maximum absolute error to ridge maximum absolute error is 1.4673, and no task fails a declared stability screen. All 5,376 task–bank–stage diagnostic records report zero affected rows for coordinate, gate, kernel, KAHM, and RBF query safeguards. No local model uses a rank-zero fallback, and no unexpected factor failure occurs. Counts refer to unions over local models within a task, bank and stage; the same query would count again if a different bank or stage invoked a guard. No method fails completely and no final prediction cell requires recovery. These checks cover every specified task and cap candidate.

Table 12: C-selected correction activation. Nonlinear and linear counts have 432 and 144 task/horizon cells, respectively. The four mixing-weight columns combine all 576 cells. A policy choice of raw ridge is counted as zero correction; raw ridge itself has no mixing selection. Nonzero mixing is not an empirical guarantee of D improvement, and no minimum activation frequency was part of the decision.

Procedure	NL active	Linear active	$\gamma = 0$	1/4	1/2	1
Raw RBF	432	56	88	53	147	288
Boosted trees	416	43	117	104	189	166
Persistence	52	20	504	71	1	0
Common features	431	51	94	14	52	416
Bounded KAHM departure	431	52	93	21	73	389
Bounded RBF departure	430	53	93	21	62	400
Unbounded KAHM departure	432	51	93	26	98	359
Unbounded RBF departure	432	48	96	23	68	389
Selected non-KAHM	431	72	73	38	99	366

Table 13: Whole-procedure choices made by the C-selected non-KAHM policy. Each task contributes one choice across all horizons after each candidate's own horizon-wise C mixing selection. Both KAHM arms are excluded. These are validation choices, not choices made from D errors.

Law	η	Ridge	Raw RBF	Boosting	Persist.	Common	RBF	RBF unb.
Nonlinear AR	0.0	0	0	0	0	8	3	13
Nonlinear AR	0.1	0	0	0	0	6	6	12
Van der Pol	0.0	0	0	24	0	0	0	0
Van der Pol	0.1	0	0	7	0	0	16	1
Duffing	0.0	0	2	1	0	0	2	19
Duffing	0.1	0	8	2	0	3	6	5
Linear AR	0.0	0	7	5	5	1	3	3
Linear AR	0.1	0	6	6	4	1	1	6

B.6 Measured costs and reproducibility limits

Each task runs in a fresh process with one assigned CPU and one numerical thread; up to three tasks run concurrently on distinct CPUs. The charged primitive-node costs exclude diagnostic-only rank calculations; full-suite timings, including diagnostics, are retained separately. Node accounting includes model candidates that are actually evaluated, rather than only the winning hyperparameters. Wall and CPU times depend on the recorded machine and software environment. The peak whole-task process RSS is 162.9 MiB; it includes imports, all candidates, shared diagnostics and evidence arrays, and is not a per-method RAM estimate. Serialized state and owned array buffers omit transient workspace and do not measure online update latency. The complete fit stage takes 308.2 seconds and the prediction stage 227.5 seconds on this shared host, including worker and evidence overhead. They are not per-method latency estimates. The study imposes no application-specific cost ceiling.

The analysis program verifies the frozen source and staged input checksums, the complete task roster, exact target correspondence, A variance scales, policy predictions, nonfinite/failure accounting and all primary aggregation. An independent post-release audit regenerates episodes from the pinned generators and

Table 14: Mean charged resources over all 192 equal-sized task suites. Fitting and selection include required A fitting, B selection and C calibration. Dimension-selected procedures pay for all six native cap candidates. Shared computation nodes are charged once to every procedure requiring them; policy fitting pays the union of its non-KAHM candidates. State denotes the pruned selected deployment object, while arrays count its owned NumPy buffers. The final two rows give fixed-cap KAHM controls. These are descriptive implementation costs, not isolated speed benchmarks or minimum deployment complexity.

Procedure	Wall (s)	CPU (s)	State (KiB)	Arrays (KiB)
Raw ridge	0.001	0.001	1.1	0.6
Raw RBF	0.015	0.015	68.1	66.7
Boosted trees	0.681	0.681	370.5	332.1
Persistence	0.001	0.001	1.2	0.7
Common features	0.187	0.187	17.3	11.7
Bounded KAHM departure	0.490	0.490	136.9	129.5
Bounded RBF departure	0.380	0.380	68.7	61.9
Unbounded KAHM departure	0.484	0.484	136.7	129.2
Unbounded RBF departure	0.377	0.377	68.2	61.3
Selected non-KAHM	1.261	1.261	136.9	123.7
KAHM, fixed cap 2	0.085	0.086	122.1	114.7
KAHM, native cap 20	0.101	0.101	183.3	175.8

independently checks seeds, noise, windows, targets, B and C choices, local bases, and all 27 numerical tables. The manuscript builder verifies the analysis table hashes before producing these displays.

The frozen source inventory has 151 files. Model states are hashed before and after prediction, all target-bearing files have explicit release checks, and every candidate forecast is retained. Six disjoint numerical archives group repetitions 0–3, 4–7, 8–11, 12–15, 16–19, and 20–23; the first also contains global inventories and seals. Their manifest records every member checksum. The archived forecast arrays define the reported scores. Re-evaluating serialized states additionally requires compatible numerical kernels and package versions; processor-dependent vectorized arithmetic can change individual floating-point values.

Numerical development checks, model replay and independent calculations are consistency checks; they do not constitute a new scientific replication. The protocol was specified locally before evaluation and was informed by the earlier experiments. It was not publicly preregistered. Neither these known-law repetitions nor the algebraic distinctness certificate establishes a natural-application forecasting claim.